

Choose the vision task

Before choosing a model, decide what information must enter the system, what must leave it, and what counts as useful.

15–20 MINUTES

No ML prerequisites

BROWSER EXERCISE

1 • Observation

What the camera measures.

2 • Task

Classification, detection, or segmentation.

3 • Output

The structured answer a consumer needs.

4 • Evidence

A test tied to the real use.

The question

“Recognize colored cubes” sounds precise, but it hides several different problems. Does the robot need to know whether any cube exists? Which colors are visible? Where each cube is? Which pixels belong to it? A good vision project begins by turning the informal goal into an **input/output contract**.

Central mental model

A vision system maps an observation to a useful representation: *image* → *answer*. The required answer determines the task, data, model, and evaluation.

Three common vision tasks

Image classification assigns one or more labels to an entire image. “This image contains a red cube” is classification. It does not say where the cube is.

Object detection returns a collection of objects. Each detection normally includes a class, a bounding box, and a score. It answers both *what* and approximately *where*.

Semantic segmentation assigns a class to each pixel. It can describe an object's outline, but its annotation and output are denser than a box.

Need	Suitable task	Typical output
Choose one label for a photo	Classification	red_cube
Find every cube	Detection	class + box + score
Know the exact visible outline	Segmentation	class per pixel

A contract has four parts

1. **Input:** what observation is available, including camera and timing assumptions.
2. **Output:** what structured information the next component receives.
3. **Operating conditions:** lighting, viewpoints, distances, backgrounds, and hardware.
4. **Success criterion:** a measurable condition connected to the robot's job.

“High accuracy” is incomplete. A detector that misses the cube the robot is about to grasp may be unacceptable even if it performs well on many easy images. Metrics become meaningful only after the use is clear.

Browser exercise: choose the smallest sufficient task

A fixed camera watches a conveyor. The controller only needs to stop when *any red cube* appears; it does not need its location.

Reveal your task choice

Now change one requirement: a robot arm must point at every red cube. Which task becomes necessary?

Reveal the changed answer

Applied to our cubes

REPOSITORY OBSERVATION The project definition asks for three color classes and locations that other robot components can consume. The node publishes detection messages containing class, score, and box. That makes object detection a defensible task choice.

OPEN DESIGN QUESTION A bounding box may be enough for recognition and visualization, but grasping may later require depth, a 3-D point, or pose. That future need should extend the output contract rather than be silently assumed today.

Project anchors: project definition and architecture.

Check your understanding

Which task directly answers “what and where” for multiple objects?

- ☐ Image classification only
- ☐ Object detection task
- ☐ Pixel normalization step

Explain it back

Without looking up: explain why “recognize colored cubes” is not yet a complete engineering specification. Then give one input, one output, one operating condition, and one success criterion for this project.

Primary sources

- Szeliski, *Computer Vision: Algorithms and Applications* — introduction and recognition task framing.
- Microsoft COCO paper — detection and segmentation tasks in complex scenes.